MIL-C-60556 (MU)
7 November 1966
SUPERSEDING
FA-PD-MI-2223, Rev 1
2 June 1959

MILITARY SPECIFICATION
CARTRIDGE, INITIATOR, M68, ASSEMBLY

1. SCOPE

1.1 This specification covers Cartridge, Initiator, M68, for use in conjunction with propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS
Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-C-60539	- Coatings, Anodic, Conventional, for Aluminum and Aluminum Alloys

STANDARDS
Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber O-rings
MIL-STD-639	- Surface Roughness Values Preferred for Army Ordnance Material
MIL-STD-1167	- Ammunition Data Card
MS28741	- Hose Assembly, Detachable End-Fitting, Medium Pressure

FSC 1377

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:43Z
Check the source to verify that this is the current version before use.

2.1 (cont)

DRAWINGS

Munitions Command

D8593492	-	Cartridge, Initiator, M68 Assembly
F8797454	-	Box, Packing, Ammunition for Cartridge, Initiator M68 or T236
F8797450	-	Carton, Packing, Ammunition for Cartridge, Initiator M68 or T236
IEL-8593492	-	Index of Inspection Equipment List for Cartridge, Initiator, M68

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 First article sample.- Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.3 Vibration.- The assembled cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer, igniter or propellant, or derangement of components.

3.4 Protective finish.- Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of configuration.

3.5 Functional.

3.5.1 The cartridge shall not fail to fire.

3.5.2 Ballistics.- The cartridge when tested in an M8 Initiator and temperature conditioned in accordance with table I, shall produce the pressures specified therein at the end of 30 feet of aircraft hose, size No. 4, MS-28741, in a 1-cubic-inch pressure station.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:43Z
Check the source to verify that this is the current version before use.

3.5.2 (cont)

Table I

Temperature	Sample	(See 6.1) Ignition Delay	Pressure (psi) Min	Max
-65° to -70° F	10	65 milliseconds (max)	1000	2200
65° to 75° F	10	65 milliseconds (max)	1200	2300
160° to 165° F	10	65 milliseconds (max)	1300	2400

3.6 Workmanship.- Workmanship shall be in accordance with the requirements of the applicable drawings, specifications and the following:

3.6.1 Metal defects.- The metal parts assembly shall be free from cracks, splits, cold shuts, inclusions, porosities or other metal defects.

3.6.2 Burr.- The part or assembly shall not have a burr which might interfere with assembly or function of the item, or which might be injurious to personnel handling the cartridge.

3.6.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 35 assembled cartridges, 50 unplated cartridge cases for hardness test, 10 sets of unassembled components with explosive components for visual and dimensional examination and 23 sets of unassembled components without explosive components for protective finish tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:43Z
Check the source to verify that this is the current version before use.

4.2.1 (cont)

Table II

Paragraph Requirement	Paragraph Test	Test	Sample Size	Type of Test Destr	Type of Test Nondestr
3.3	4.4.2	Vibration	12 (from Ballistic Sample)	-	x
Dwg	4.4.3	Airtightness	Each	-	x
3.5.2	4.4.5.1	Ballistics	30	x	-
Dwg	4.3.3.1	Hardness	50 (Unplated)	x	-
Dwg	4.4.5.2	Sealant	5	x	-
3.4	4.3.3.4	Protective Finish	23	x	-

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials (including packaging and packing) shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics.- The sample for ballistic tests shall be temperature conditioned and tested to determine compliance with table I. The sample for firing shall consist of the following:

Table III

Temperature	Sample	Vibrated
-65° to -70° F	6	4
65° to 75° F	6	4
160° to 165° F	6	4

4.2.1.3 Initial production sample failure.- Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:43Z
Check the source to verify that this is the current version before use.

4.3.1.2 Size of lot.- Unless otherwise specified by the procuring agency, an ammunition lot shall consist of 1,200 cartridges, maximum, exclusive of test samples and shall consist of:

a. Components produced by a homogeneous process.

b. Propellant from not more than one lot.

c. Igniter from not more than one lot.

d. Primers from not more than one lot.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis, in accordance with the classification of defects, using the sampling plans and criteria in tables IV and V. For an individual defect (single characteristic) the criteria in table IV shall be used. For total defects within a major or minor group, the criteria in table V shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter. The Ordnance Standard Finish Blocks prescribed in MIL-STD-639 shall be used as a basis of comparison for surface roughness determination. Inspection for rubber O-rings shall be in accordance with MIL-STD-413.

Table IV

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% examination	-	-	-	-
301 - 1,300	110	1	2	2	3
1,301 - 3,200	150	2	3	3	4

Table V

Total Allowable Defects per Sample Size

Sample Size for Major Defects . .	110	150
Number of Defects in Class		
2 - 3	1	2
4 - 7	2	3
8 - 19	3	4
20 - 43	4	5
Over 43	6	8

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:43Z
Check the source to verify that this is the current version before use.

4.3.2 (cont)

Sample Size for Minor Defects . .	110	150
Number of Defects in Class		
2 - 3	2	3
4 - 7	3	4
8 - 16	4	5
17 - 48	6	8
Over 48	8	11

4.3.2.1 Cartridge Case, Dwg 8593493 covering a detail of Dwg D8593492.

Defect	Method of Inspection
CRITICAL	
1. Thickness of base, min	Gage
2. Thickness of wall, min	Gage
3. Metal defective	Visual
MAJOR	
101. Concentricity of cartridge case head with body	Gage
102. Straightness of base	Gage
103. Diameter at datum location	Gage
104. Protective finish	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
201. Diameter at base	Gage
202. Total length	Gage
203. Thickness of wall, max	Gage
204. Thickness of base, max	Gage
205. Radius missing or incorrect	Visual
206. Surface finish improper	Visual

4.3.2.2 Sealing Disc, Dwg C8593494 covering a detail of Dwg D8593492.

CRITICAL

None defined.

MAJOR

101. Diameter, min	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:43Z
Check the source to verify that this is the current version before use.

4.3.2.2 (cont)

	Defect	Method of Inspection
	102. Thickness (after varnish)	Gage
	103. Metal defecteve	Visual
	104. Protective finish inadequate or defective	Visual
MINOR		
	201. Diameter, max	Gage
	202. Varnish missing or inadequate	Visual
	203. Marking missing, incorrect or illegible	Visual

4.3.2.3 Cartridge Head, Dwg C8593495 covering a detail of Dwg D8593492.

CRITICAL		
	1. Depth of primer pocket	Gage
	2. Metal defective	Visual
MAJOR		
	101. Concentricity of large body with primer pocket	Gage
	102. Diameter of large body, min	Gage
	103. Diameter of primer pocket	Gage
	104. Location of datum diameter	Gage
	105. Protective finish	Visual
	106. Surface finish of primer pocket and face of head	Visual
	107. Protective finish	Visual
	108. Burr, except as permitted	Visual
	109. Foreign matter	Visual
MINOR		
	201. Diameter of large body, max	Gage
	202. Diameter of flash hole	Gage
	203. Diameter of small end	Gage
	204. Total length	Gage
	205. Radius or chamfer missing or incorrect	Visual
	206. Surface finish improper, except as otherwise classified	Visual

4.3.2.4 Cartridge, Initiator, M68 Assembly, Dwg D8593492.

CRITICAL		
	1. Primer not flush or below cartridge head	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:43Z
Check the source to verify that this is the current version before use.

4.3.2.4 (cont)

Defect	Method of Inspection
2. Primer not fully seated	X-ray/Visual 1/
3. Primer pellet missing	X-ray/Visual 1/
4. Anvil missing, cracked, broken or inverted	X-ray/Visual 1/
5. Any part missing	X-ray and Visual 1/
6. Improper assembly	X-ray and Visual 1/
7. Mixed ammunition	Visual
8. Metal defects	Visual
9. Crimp missing or incorrect	Visual
10. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Concentricity of head with body	Gage
102. Diameter of body, max	Gage
103. Diameter of head, max	Gage
104. Length of head	Gage
105. Location of datum diameter	Gage
106. 0-ring split, cracked or cut	Visual
107. Burr	Visual
108. Foreign matter	Visual
MINOR	
201. Total length, max	Gage
202. Varnish missing or inadequate	Visual
203. Sealant missing or improperly applied	Visual

1/ The assembled cartridge shall be x-rayed with the axis of the cartridge perpendicular to the rays of the x-ray machine. The x-ray negatives, properly identified, shall be forwarded to the procuring activity.

4.3.2.5 Carton, Packing, Ammunition for Cartridge, Initiator M68 or T236, Dwg F8797450.

CRITICAL
None defined.

MAJOR
None defined.

MINOR

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:43Z
Check the source to verify that this is the current version before use.

4.3.2.5 (cont)

	Defect	Method of Inspection
201.	Contents loose	Manual and Visual
202.	Quantity of contents incorrect	Visual
203.	Separator or filler missing	Visual
204.	Heat seal incomplete, loose or distorted	Visual
205.	Cut or damaged through impregnated box	Visual
206.	Marking missing, incorrect or illegible	Visual

4.3.2.6 Box, Packing, Ammunition for Cartridge, Initiator M68 or T236, Dwg F8797450.

CRITICAL
None defined.

MAJOR
None defined.

MINOR

201.	Contents loose	Manual and Visual
202.	Hardware missing, loose, broken or improperly engaged	Visual
203.	Strapping missing, loose, broken or improperly assembled	Visual
204.	Contents of box exposed	Visual
205.	Box damaged	Visual
206.	Marking missing, incorrect or illegible	Visual
207.	Number of cartons incorrect	Visual
208.	Car seal missing, unsealed or improperly positioned	Visual

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4:

4.3.3.1 Hardness.- Fifty units from each heat-treated lot of components shall be tested. Failure of any unit to comply with the requirement of the drawing shall be cause for rejection of the lot of components.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:43Z
Check the source to verify that this is the current version before use.

4.3.3.2 Vibration.- Twelve of the 30 cartridges of the initial production sample shall be subjected to vibration. The 12 cartridges shall then be x-rayed. Failure of any cartridge to comply with the requirement of 3.3 shall be cause for rejection of the lot. Any change in the process of a lot of primers in subsequent production shall be reason to select an additional sample of 12 cartridges for vibration tests.

4.3.3.3 Airtightness.- Each cartridge shall be subjected to the specified pressure differential. Failure to comply with the drawing requirements shall be cause for rejection of the cartridge.

4.3.3.4 Protective finish.- Twenty-three units of each component shall be selected for protective finish tests. Failure of any component to comply with the requirement of 3.4 shall be cause for rejection of the lot.

4.3.3.5 Functional.- The following tests shall be performed by the Government:

4.3.3.5.1 Ballistics.- Thirty assembled cartridges from the initial production sample and 30 cartridges from each subsequent production lot shall be functionally tested. Failure of any cartridge to comply with the requirement of 3.5.2 shall be cause for rejection of the lot.

4.3.3.6 Sealant application.- Five cartridges from each production lot shall be selected for this test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the lot of cartridges.

4.3.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-8593492.

4.4 Test methods and procedures.

4.4.1 Hardness.- A hardness testing machine shall be used for determining the hardness values specified on the applicable drawing.

4.4.2 Vibration.- The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.3 Airtightness.- The method for determining airtightness shall be by use of equipment listed on IEL-8593492.

4.4.4 Protective finish.- Test procedures for protective finish shall be in accordance with MIL-C-60539. The cartridge case shall be sectioned longitudinally to better expose the significant surfaces to the salt spray.

Source: https://assist.dla.mil -- Downloaded: 20261O-23T00:43Z
Check the source to verify that this is the currentversion before use.

4.4.5 Functional.

4.4.5.1 Ballistics.- The cartridge shall be loaded into an acceptable M8 Initiator, divided into three equal groups and temperature conditioned in accordance with table I. The initiator shall have end caps to cover the gas port during the conditioning period. The cartridge shall be tested within 10 minutes after removal from the conditioning chamber. When this time is exceeded, the initiator shall be re-conditioned in accordance with 4.4.5.1.1.

4.4.5.1.1 Temperature conditioning.- A standard M8 Initiator loaded with a dummy M68 Cartridge, containing a temperature sensitive element, shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

4.4.5.2 Sealant application.- The cartridge shall be disassembled in such a manner that determination can be made that the sealant has been applied for the full length of the joining surfaces at the junction of the sealing disc and case sidewall and that the propellant cavity is free of sealant.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking - Level A.- Packing and marking shall be in accordance with drawings F8797450 and F8797454.

5.2 Data cards.- Data card information shall be in accordance with MIL-STD-1167.

6. NOTES

6.1 Ignition delay is defined as the time that elapses between the firing of the primer and the beginning of a continuous pressure rise on a pressure time record.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1377-A062

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:43Z
Check the source to verify that this is the current version before use.

To detach this form, cut along this line

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:43Z
Check the source to verify that this is the current version before use.